

Principal of Math Analysis

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Chapter 1

Real and Complex Number Systems

1.1 Introduction

Example 1.1.1 (An introductory example to the problem with rational numbers)

$$p^2 = 2$$

cannot be satisfied by any rational number $p \in \mathbb{Q}$,

Proof: **Proof by contradiction:** If there is such p , let

$$p = \frac{a}{b} \tag{1.1}$$

where $a, b \in \mathbb{Z}$, both even. Next,

$$m^2 = 2n^2 \tag{1.2}$$

Since $2n^2$ is even, m^2 is even and m are **not both even**, and so m^2 is also divisible by 4. It follows that right hand side is divisible by 4, making n^2 even, implying n is even. The assumption that 1.1 indicates that m and n are **both even**, which contradicts to our choice of m and n . Thus, 1.1 doesn't stand for $p \in \mathbb{Q}$. ☹

Example 1.1.2 (Looking more closely)

Let

- A the set of all positive rational p such that $p^2 < 2$, which contains **no largest**, meaning for every $p \in A$ there is a rational $q \in A$ s.t. $p > q$
- B the set of all positive rational p such that $p^2 > 2$, which contains **no smallest**, meaning for every $p \in B$ there is a rational $q \in B$ s.t. $q > p$

Proof: We associate with each rational $p > 0$ the number

$$q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} \tag{1.3}$$

Then

$$\begin{aligned}
 q^2 - 2 &= \frac{(2p+2)^2}{(p+2)^2} - 2 \\
 &= \frac{4p^2 + 8p + 4}{(p+2)^2} - 2 \\
 &= \frac{4p^2 + 8p + 4 - 2p^2 - 8p - 8}{(p+2)^2} \\
 &= \frac{2p^2 - 4}{(p+2)^2} \\
 &= \frac{2(p^2 - 2)}{(p+2)^2}
 \end{aligned} \tag{1.4}$$

If

- p is in A , since $p^2 < 2 \Rightarrow p^2 - 2 < 0$, 1.3 shows that $q > p$ because

$$\frac{p^2 - 2}{p + 2} < 0 \Rightarrow p - \frac{p^2 - 2}{p + 2} > p$$

,and 1.4 shows that $q^2 < 2$. Thus $q \in A$

- p is in B , since $p^2 > 2 \Rightarrow p^2 - 2 > 0$, by 1.3 shows that $0 < q$. Moreover, $\frac{p^2 - 2}{p + 2} > 0$ and by 1.3, $q < p$. 1.4 shows that $q^2 > 2$ since the right side is positive. Thus $q \in B$. $\odot$

1.2 Ordered sets

Definition 1.2.1: Ordered set

Let S be a set, and order on s is a relation, denoted by $<$, with the following properties:

1. If $x \in S$ and $y \in S$ then one and only one of the statements

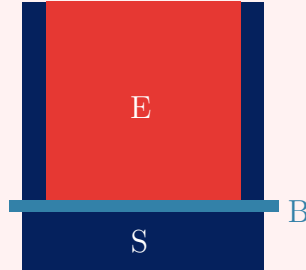
$$x < y, \quad x = y, \quad y < x$$

is true.

2. If $x, y, z \in S$, if $x < y$ and $y < z$, then $x < z$. The statement " $x < y$ " may be read as " x is less than y " or " x is smaller than y " or " x precedes y ".

Definition 1.2.2: Lower bound

Suppose S is an ordered set, and $E \subset S$. There is $\beta \in S$ such that $x \leq \beta$ for every $x \in E$, then we say that E is **bounded above** and call β and **upper bound** of E . Lower bounds are defined similarly (change $\leq$ to $\geq$). Here is an illustration of lower bound:



Lower bound

Where E is a subset bounded below an ordered set S , and B is the lower bound.

1. sup is least upper bound or LUB
2. inf is greatest lower bound or GLB

Note:-

Both LUB and GLB can be **either in or not in the set**.

Definition 1.2.3: Least Upper-bound Property

Given an ordered set S . If:

1. $E \subset S$
2. E isn't empty
3. E is bounded above

then S is said to have the *least-upper-bound-property*

Theorem 1.2.1 LUB and GLB

Every ordered set with least-upper-bound-property also has greatest-lower-bound property

Theorem 1.2.2

Given S , an ordered set, has the *least-upper-bound-property*. $B \subset S$, and is bounded below. Let L be the set of all lower bounds of B . Then

$$\alpha = \sup L$$

exists in S , and $\alpha = \inf B$

1.3 Fields

Definition 1.3.1: Field

A field is a set F with two operations: addition and multiplication. It follows **field axioms**: A,M,D:

1. Axioms of addition (A):
 - (a) closure under addition: given $x \in F$ and $y \in F$ their sum $x + y \in F$
 - (b) addition is commutative: $x + y = y + x$
 - (c) addition is associative: $(x + y) + z = x + (y + z)$
 - (d) additive identity: defined by $0 \in F$ such that $x + 0 = x$ for every $x \in F$
 - (e) additive inverse: given $x \in F$, there exists $-x \in F$ such that $x + (-x) = 0$
2. Axioms of multiplication (M):
 - (a) closure under addition: given $x \in F$ and $y \in F$ their product $x \cdot y \in F$
 - (b) multiplication is commutative: $x \cdot y = y \cdot x$
 - (c) multiplication is associative: $(x \cdot y) \cdot z = x \cdot (y \cdot z)$
 - (d) multiplicative identity: denoted by 1, such that $x \cdot 1 = x$ for every $x \in F$
 - (e) multiplicative inverse: given $x \in F$ and $x \neq 0$, there exists $x^{-1} \in F$ such that $x \cdot x^{-1} = 1$
3. The Distributive law (D):

$$x(y + z) = xy + xz$$

holds for all $x, y, z \in F$.

Proposition 1.3.1

According to the axioms of addition (A), it's implied that:

1. If $x + y = x + z$, then $y = z$

Proof:

$$\begin{aligned} y &= 0 + y = (-x + x) + y = -x + (x + y) \\ &= -x + (x + z) = (-x + x) + z = 0 + z = z \end{aligned}$$



2. If $x + y = x$ then $y = 0$

Proof: Continuing from previous proof, take $z = 0$:

$$x + y = x + 0 = x + z \Rightarrow y = 0$$



3. If $x + y = 0$, then $y = -x$

Proof:

$$\begin{aligned} x + y &= x + (-x) \\ x - x &= (-x) - y \\ 0 &= -x - y \\ y &= -x \end{aligned}$$



$$4. -(-x) = x$$

Proof: Since $-x + x = 0$, by 3, $x = -(-x)$



Proposition 1.3.2

According to the axioms of multiplication (M), (the proofs are similar to above), it's implied that:

1. If $x \neq 0$ and $xy = xz$ then $y = z$

Proof:

$$\begin{aligned} y &= (x \frac{1}{x})y \\ &= (xy) \frac{1}{x} \\ &= (xz) \frac{1}{x} \\ &= (x \frac{1}{x})z \\ &= z \end{aligned}$$



Note that $x \neq 0$ might a hint for the proof: the proof might involve x as a denominator so that it needs to be non-zero.

2. If $x \neq 0$ and $xy = x$ then $y = 1$

Proof:

$$\begin{aligned} y &= y \cdot 1 \\ &= y(x \frac{1}{x}) \\ &= (xy) \frac{1}{x} \\ &= x \frac{1}{x} \\ &= 1 \end{aligned}$$



This is very similar to the proof above, both using the technique of **construction**: use 1 to create $x \cdot \frac{1}{x}$.

3. If $x \neq 0$ and $xy = 1$ then $y = 1/x$

Proof: Adjust the previous proof by replacing $xy = 1$ into

$$\begin{aligned} y &= (xy) \frac{1}{x} \\ &= 1 \cdot \frac{1}{x} \\ &= \frac{1}{x} \end{aligned}$$



4. If $x \neq 0$ then $\frac{1}{1/x} = x$

Proof:

$$\begin{aligned} 1 \cdot \frac{1}{1/x} &= \left(x \frac{1}{x}\right) \frac{1}{1/x} \\ &= x \left(\frac{1}{x} \frac{1}{1/x}\right) \\ &= x \cdot 1 \\ &= x \end{aligned}$$



This is still construction

Proposition 1.3.3

The field axioms imply the following statements, for any $x, y, z \in F$

1. $0x = 0$

Proof:

$$0x + 0x = (0 + 0)x = 0x$$

Since proven above (2), taking $0x = x$ and $0x = y$, we have

$$0x = 0$$



2. If $x \neq 0$ and $y \neq 0$ then $xy \neq 0$

Proof: This is a proof by contradiction: if $x \neq 0$ and $y \neq 0$ but $xy = 0$, then proof (1) gives shows that $y = 0$, which contradicts to the assumption.

The proof also gives:

$$1 = \left(\frac{1}{y}\right) \left(\frac{1}{x}\right) xy = \left(\frac{1}{y}\right) \left(\frac{1}{x}\right) 0 = 0$$

which is a proof by contradiction (from Rudin)



3. $(-x)y = -(xy) = x(-y)$

Proof:

$$(-x)y + xy = (-x + x)y = 0y = 0$$

by the additive inverse property 3, we have

$$(-x)y = -(xy)$$

Similarly, since

$$x(-y) + xy = x(-y + y) = 0x = 0$$

we can prove the other half: $-xy = x(-y)$



4. $(-x)(-y) = xy$

Proof: By previous proof 3,

$$(-x)(-y) = -[x(-y)] = -[-(xy)]$$

and by the previous proposition proven 4, $-[-(xy)] = xy$



Definition 1.3.2: Ordered field

An **ordered field** is a field F which is also an **ordered set**, such that

1. $x + y < x + z$ if $x, y, z \in F$ and $y < z$ as known as **trichotomy** property
2. $xy > 0$ if $x \in F, y \in F, x > 0$ and $y > 0$

This sounds pretty intuitive for the fields we know, e.g. $\mathbb{R}$ and $\mathbb{Q}$. However, there are some fields that are not ordered, e.g. $\mathbb{C}$.

Proof: By trichotomy, either $i > 0$ or $i < 0$ (since $i \neq 0$) Considering two cases regarding $i \in \mathbb{C}$:

1. if i is positive, then by (2) $i^2 > 0$ since $i > 0$ by (2) $i^2 > 0$, which gives $i^2 = -1 > 0$, and it's contradictory
2. if i is negative, then by (2) $(-i)$ is positive, and by (2) $(-i)^2 > 0$.

$$(-i)^2 = (-1 \cdot i)(-1 \cdot i) = (-1)^2 \cdot i^2 = -1 > 0$$

which is also contradictory

Hence no order on $\mathbb{C}$ can satisfy the axioms, so $\mathbb{C}$ is not an ordered field. ☹

Proposition 1.3.4 Ordered field properties

1. If $x > 0$ then $-x < 0$, and vice versa

Proof: If $x > 0$, then $0 = -x + x > -x + 0$, so that $-x < 0$.

If $x < 0$, then $0 = -x + x < -x + 0$, so that $-x > 0$ ☺

2. If $x > 0$ and $y < z$ then $xy < xz$.

Proof: Since $z > y$, we have $z - y > y - y = 0$, hence $x(z - y) > 0$, and therefore

$$xz = x(z - y) + xy > 0 + xy = xy$$

☺

3. If $x < 0$ and $y < z$ then $xy > xz$

Proof: By (1), (2) and prop 3,

$$-[x(z - y)] = (-x)(z - y) > 0$$

so that $x(z - y) < 0$, hence $xz < xy$ ☺

4. If $x \neq 0$ then $x^2 > 0$. In particular, $1 > 0$

Proof: If $x > 0$, (2) of ordered field definition 1.3 gives $x \cdot x = x^2 > 0$.

If $x < 0$, $-x > 0$, hence $(-x)^2 > 0$. By prop 4, $x^2 = (-x)^2$.

Since $1 = 1^2, 1 > 0$ ☺

5. If $0 < x < y$ then $0 < \frac{1}{y} < \frac{1}{x}$

Proof: If $y > 0$ and $v \leq 0$, then $yv \leq 0$. On the other hand, $y \cdot (\frac{1}{y}) = 1 > 0$. Hence $\frac{1}{y} > 0$. Likewise, $\frac{1}{x} > 0$.

Given that $\frac{1}{x} \cdot \frac{1}{y} = \frac{1}{xy} > 0$, and $x < y$, multiplying both sides by $\frac{1}{xy}$ gives

$$\begin{aligned} \frac{1}{xy} \cdot x &< \frac{1}{xy} \cdot y \\ \frac{1}{y} &< \frac{1}{x} \end{aligned}$$

1.4 The real field

Theorem 1.4.1

There exists an ordered field $\mathbb{R}$ which has the least-upper-bound-property. Moreover $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield. (Since the proof is long and tedious, skipped.)

Theorem 1.4.2

1. (Archimedian property of $\mathbb{R}$) If $x \in \mathbb{R}, y \in \mathbb{R}$ and $x > 0$, there is a positive integer n such that

$$nx > y$$

2. ($\mathbb{Q}$ is dense in $\mathbb{R}$) If $x \in \mathbb{R}, y \in \mathbb{R}$ and $x < y$, then there exists a $p \in \mathbb{Q}$ such that $x < p < y$.

Proof: 1. Let A be the set of nx , where n runs through all positive integers. If 1) is false, then y would be an upper bound of A . But then A has a least upper bound in $\mathbb{R}$. Let $a = \sup A$, since $x > 0$, $a - x < a$, and $a - x$ is not an upper bound of A . Hence $a - x < mx$ for some positive integer m . But then $a < (m + 1)x \in A$, which is impossible, since a is an upper bound of A .

2. Since $x < y$, we have $y - x > 0$, and 1) furnishes a positive integer n such that

$$n(y - x) > 1 \Rightarrow ny > 1 + nx$$

Apply 1) again, to obtain positive integers m_1, m_2 such that $m_1 \cdot 1 > nx, m_2 \cdot 1 > -nx$. Then

$$-m_2 < nx < m_1$$

hence there is an integer m with $-m_2 \leq m \leq m_1$ such that

$$m - 1 \leq nx \leq m$$

combining the inequalities

$$nx < m \leq 1 + nx < ny$$

since $n > 0$,

$$x < \frac{m}{n} < y$$

this proves 2), with $p = m/n$.

Theorem 1.4.3 1.21

For every real $x > 0$ and every integer $n > 0$, there is one and only one positive real y such that $y^n = x$. This number y is written $\sqrt[n]{x}$ or $x^{1/n}$.

Proof: That there is at most one such y is clear, since $0 < y_1 < y_2$ implies $y_1^n < y_2^n$.

Let E be the set consisting of all positive real numbers t such that $t^n < x$. If $t = x/(1+x)$ then $0 \leq t < 1$. Hence $t^n \leq t < x$. Thus $t \in E$, and E is not empty.

If $t > 1+x$ then $t^n \geq t > x$, so $t \notin E$. Thus $1+x$ is an upper bound of E . Hence Theorem 1.19 implies the existence of

$$y = \sup E.$$

To prove that $y^n = x$ we will show that each of the inequalities $y^n < x$ and $y^n > x$ leads to a contradiction.

The identity

$$b^n - a^n = (b-a)(b^{n-1} + b^{n-2}a + \cdots + a^{n-1})$$

yields the inequality

$$b^n - a^n < (b-a)nb^{n-1}$$

when $0 < a < b$.

Assume $y^n < x$. Choose h so that $0 < h < 1$ and

$$h < \frac{x - y^n}{n(y+1)^{n-1}}.$$

Put $a = y$, $b = y + h$. Then

$$(y+h)^n - y^n < hn(y+h)^{n-1} < hn(y+1)^{n-1} < x - y^n.$$

Thus $(y+h)^n < x$ and $y+h \in E$. Since $y+h > y$, this contradicts the fact that y is an upper bound of E .

Assume $y^n > x$. Put

$$k = \frac{y^n - x}{ny^{n-1}}.$$

Then $0 < k < y$. If $t \geq y - k$, we conclude that

$$y^n - t^n \leq y^n - (y-k)^n < kny^{n-1} = y^n - x.$$

Thus $t^n > x$ and $t \notin E$. It follows that $y-k$ is an upper bound of E . But $y-k < y$, which contradicts the fact that y is the *least* upper bound of E .

Hence $y^n = x$, and the proof is complete. ☺

Corollary 1.4.1

If a and b are positive real numbers and n is a positive integer, then

$$(ab)^{1/n} = a^{1/n}b^{1/n}.$$

Proof: Put $\alpha = a^{1/n}$ and $\beta = b^{1/n}$. Then

$$ab = \alpha^n \beta^n = (\alpha\beta)^n,$$

since multiplication is commutative. By the uniqueness assertion of Theorem 1.21, it follows that

$$(ab)^{1/n} = \alpha\beta = a^{1/n}b^{1/n}.$$

